Allan Wang

+1 (778)-798-1233 | allan357161@gmail.com | Personal Website | GitHub | LinkedIn | Canadian PR

EDUCATION

University of British Columbia – Bachelor of Science in Computer Science + Statistics **Sep 2022 – Nov 2027**
Cumulative GPA**: 4.33/4.33** | Relevant Courses: Computer Systems, DSA, Statistical Modelling, Relational DB

EXPERIENCES

Amazon Web Services (AWS) | Toronto, ON, Canada **Sep 2026 – Dec 2026**
Software Development Engineer Intern – Aurora Open-Source Engines *TypeScript, AWS, React.js, PostgreSQL*

- Designed system architecture and implemented the dashboard TypeScript & React.js for analyzing RDS traces, replacing old scripts to format machine-optimized data and improving analysis overhead for on-call duties.
- Enabled multiple routes to ingest and wrangle data equivalently to render visualizations gracefully.

Rivian and Volkswagen Group Technologies | Vancouver, BC, Canada **Sep 2025 – Apr 2026**
Software Engineer Intern – Infotainment Platform *C++, Embedded Linux, GenAI*

- Developed infotainment platform service for centralized performance metrics monitoring; processed metrics into summary KPIs and reported to telemetry across multiple Virtual Machines.
- Implemented proc-based system-wide and per-process CPU, memory, and I/O scanners, reducing metric collection cycle latency by **96% (15s → 0.6s),** bypassing delays caused by vendor API limitations.
- Refactored scanners via a registry singleton and base scanner strategy, reducing ~200 lines of code/scanner.
- Designed unit tests for scanners, service overhead tests, and validation tests to ensure service outputs matched built-in Linux command measurements within a **2%** deviation.
- Resolved SELinux denial flooding in logcat by configuring read permissions, reducing **>20 entries/second**.

LED Smart Inc. | Surrey, BC, Canada **Dec 2024 – Aug 2025**
Firmware Developer Intern – R & D *C, Zephyr RTOS, Thread IoT*

- Designed and implemented firmware on nRF54L15 using the Zephyr RTOS and OpenThread in **C**; leveraged peripherals such as **GPIO** and **UART** to achieve control over LED states as a robust Thread network.
- Refined Thread device commissioning process by applying mutex locking and isolated blocking functions with multi-threading, reducing main thread delay by **1500ms** and improving system responsiveness during boot-up.
- Implemented and validated custom networking protocols using **UDP** with custom reliability mechanisms.

UBC Speed-Cubing | Vancouver, BC, Canada **Jan 2025 – Aug 2025**
Lead Software Engineer *TypeScript, Supabase, PostgreSQL, Next.js*

- Owned, designed, and developed the website for the UBC Cubing Club to be used for weekly meetings.
- Created a custom timer with reliable client-side submissions, increasing per-member solve capacity from 5 to 40 per meeting (**~8× increase**), **cut printing costs & effort entirely**, and boosted average attendance by **200%**.
- Designed tables in PostgreSQL database with Supabase and enforced Row Level Security for data integrity.

PROJECTS

Pet Management System | *Full Stack, JavaScript (React, Node, Express), SQL, Database* **Jan 2025 – Apr 2025**

- Developed a web application using **React.js with JavaScript**, that allows users to perform **CRUD** operations on a pet shelter database enabled using **custom APIs** with structured routing.
- Designed, normalized, and implemented an Oracle Database schema to optimize data storage and efficiency.

Algorithm Visualizer | *Swift, SwiftUI, Algorithms, iOS Development* **Aug 2024 – Dec 2024**

- Developed an algorithm visualizer that solves customizable mazes as an iOS application with Swift & SwiftUI.
- Implemented **BFS and DFS** to solve the maze designed by the user, rendered the solution as stepped animations.

TECHNICAL SKILLS

Programming: C, C++, JavaScript, TypeScript, Java, Python, Swift, SQL, R, HTML, CSS
Technologies: React.js, Node.js, Next.js, PostgreSQL, Java Swing, Oracle Database, IoT, Express.js
Developer Tools: Git, Repo, Postman, Docker, Vercel, JUnit, GoogleTest, Mocha, Bazel, Supabase, AWS